Worksheet 3-2 Mixed Derivative Practice (thru 3-2)
AP Calculus Name_____

Directions: Answer the problems below. Be sure to show all work ON NOTEBOOK PAPER.

1. $f(x) = 4 - 5x - \ln(x+2)$, then $f'(-1) =$

2. $g(x) = (x^2 - \sin(x))^4$, then $g'(x) =$

3. $h(x) = 2x^3 - x^2 + \frac{1}{x} - 3$, then $h'(3) =$

4. The function f is defined by $f(x) = \frac{x-1}{x+2}$. Find all points (x, y) where the line tangent to the graph of f has slope $\frac{1}{3}$.

5. What is the equation of the line tangent to the graph of $y = \ln(x^2)$ when $x = 3$?

6. If $f(x) = 4x^{-1} - \frac{x}{3} - 2$, then $f'(-3) =$

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x	-6	-2	6
$f(x)$	3	-4	5
$f'(x)$	2	1	-3

7. The table above gives selected values for the differentiable function f and its derivative f'. Let k be the function defined by $k(x) = \frac{f(3x)}{4x}$. What is the value of $k'(-2)$?

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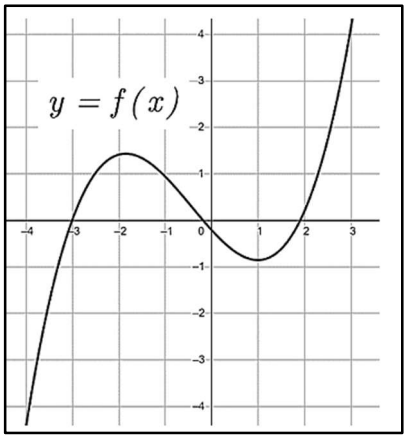
8. The graph of $f(x)$ is below. Order $f'(-3), f'(-1)$, and $f'(1)$ from least to greatest.



9. If f is the function given by $f(x) = 4 - x^3$, for which value of x is the instantaneous rate of change equal to the average rate of change of $f(x)$ on the interval $[-1,3]$?

10. $\lim_{x \to -\infty} \frac{9x - 2}{\sqrt{4x^2 - 2x + 1}} =$

11. $f(x) = (3x^2 - 5x + 1)^6$. Find $f'(x)$

12. $g(x) = \sin^2(3x - 1)$. Find $g'(x)$.

13. $g(x) = \frac{5 - 2x}{3x + 7}$, find $g'(x)$

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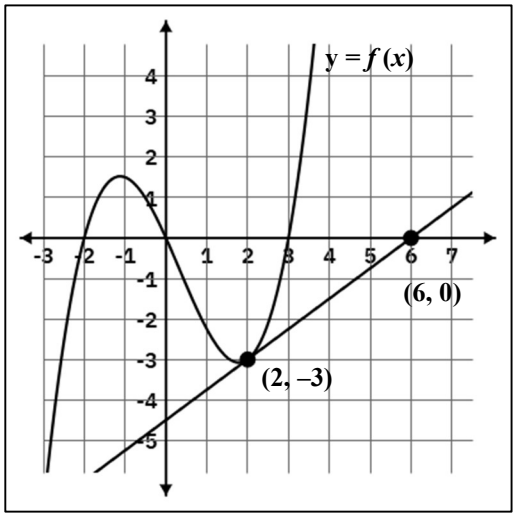
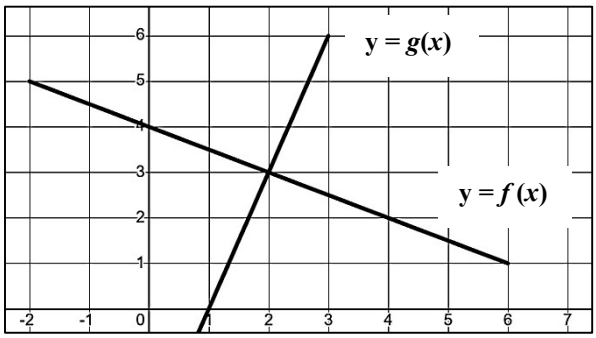
14. The graph of $f(x)$ below along with the line tangent to f at $x = 2$. Let $g(x) = \dfrac{f(x)}{e^x}$. Find $g'(2)$.



15. The figure below shows the graphs of f and g. If $k(x) = f(x)g(x)$, find $k'(2)$.



$$g(x) = \begin{cases} ax^2 + 3, & x \le 2 \\ 16x - 23, & x > 2 \end{cases}$$

16. The function g is defined above. Find the value of a that makes g continuous at $x = 2$

17. If f is the function given by $f(x) = \sin\left(\dfrac{x}{6}\right)$, write the equation of the line tangent to the graph of f at the point $\left(\pi, \dfrac{1}{2}\right)$.

18. If $f(x) = \sqrt{x^2 - 4}$ and $g(x) = 3x - 2$, find the derivative of $g\big(f(x)\big)$ at $x = 6$.

x	1	2	5	7
$f(x)$	2	8	2	-3

19. Let f be a continuous function with selected values given in the table above. Is there a time c, where $1 < c < 7$, such that $f(c) = 0$? Justify your answer.

20. $\displaystyle\lim_{x \to -3} \frac{x^2 - 9}{x^2 - 4x - 21} =$

21. Let $3x^2 - 2y + 4xy = 6$. Find $\dfrac{dy}{dx}$.

22. Write an equation of the line tangent to the curve $x^2 - 3xy + 4y = 8$ through the point $(2, -2)$.

23. Let $\sin(2x - 3y) = x^2 + 4y^3$. Find $\dfrac{dy}{dx}$.